6CCSACCM
Cooperation, Coordination and Multi-Agent Systems

Week 9

Mechanism design

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In MAS, game theory helps

- The **agents**
  - to understand what are best/reasonable/safe strategies to take.
- The **system administrator**
  - to predict possible system outcomes and make the agents to achieve stable/optimal ones.
- The **system designer**
  - to design systems with desirable outcomes.

Mechanism design

- We see that the equilibrium outcomes in some games are undesirable.
  - e.g., the Centipede game or the Prisoner's Dilemma.
- **How to design the game such that certain objectives are met in equilibrium?**
  - i.e., when each agent maximises its own (expected) utility.

Examples

- **Elections**: how to design voting protocols such that
  - a particular candidate wins or has a highest chance of winning,
  - the number of unsatisfied voters is minimised, etc.
- **Marketplaces**: how to design auctions or other trading mechanisms such that
  - the outcome is efficient (maximises the sum of utilities),
  - or maximises the seller's profit,
  - or minimises the seller's profit, etc.

Auctions

- Auctions are an interesting and important application of mechanism design.
- Auctions are protocols for multi-agent resource allocation.
  - A tool for allocating (discrete) resources/goods among selfish agents in a multi-agent system.

Auctions

The auction setting is important for two reasons:

- First, auctions are **widely used in real life**:
  - millions of people use auctions daily on Internet *consumer* Web sites to trade goods;
  - auctions have been used by *governments* around the world to sell important public resources such as access to electromagnetic spectrum;
  - all *financial markets* constitute a type of auction (one of the family of so-called *double auctions*);
  - auctions are also often used in *computational settings*, to efficiently allocate bandwidth and processing power to applications and users.

Auctions

The auction setting is important for two reasons:

- The second—and more fundamental—reason to care about auctions is that they provide a general theoretical framework for understanding **resource allocation problems** among self-interested agents:
  - formally speaking, an auction is *any* protocol that allows agents to indicate their interest in one or more resources, and that uses these indications of interest to determine both an *allocation of resources* and a *set of payments* by the agents;
  - thus, auctions are important for many computational settings (e.g., the sharing of computational power in a grid computer) that would not normally be thought of as auctions and that might not even use money as the basis of payments.

Single-good auctions

- Suppose we are selling a single indivisible good/item.
  - There is one good for sale, one seller, and multiple potential buyers.
  - Each buyer has his own (private) valuation for the good, and wishes to purchase it at the lowest possible price.
  - Such auctions are called *single-sided*, because there are multiple agents on only one side of the market.

Single-good auctions

There are a few famous families of single-good auctions:

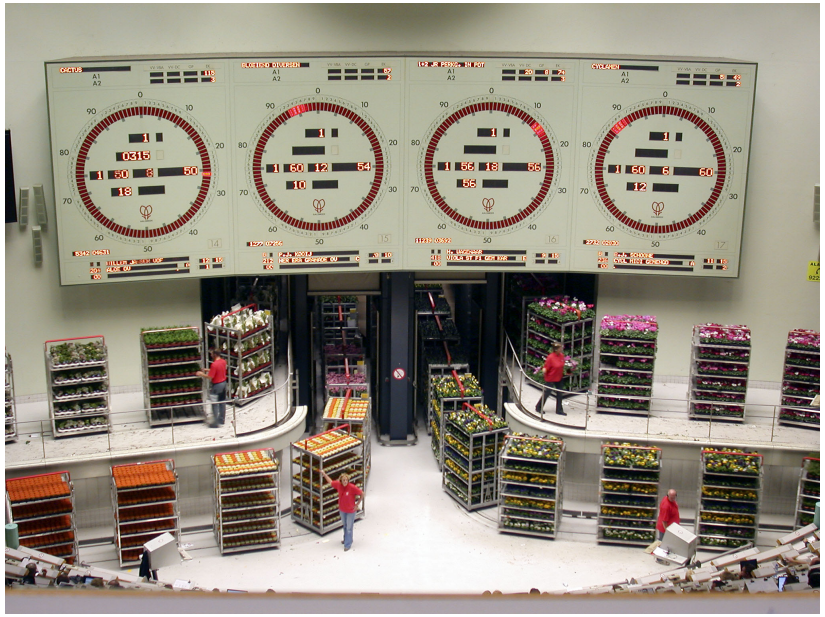
- English auction
- Dutch auction
- First-price sealed-bid auction
- Second-price sealed-bid (*aka* Vickrey) auction

Which one is better?

English auction

- Bidding starts at a low price.
- Each bidder is free to raise the price.
- Auction ends when no bidder is willing to bid higher.
- Highest bidder wins the good and pays his bid.

Dutch auction



- Bidding starts at a high price.
- The auctioneer lowers the price until someone bids.
- The good is allocated to the first bidder at that current price.

Sealed-bid auctions

- Each bidder submits his bid in a sealed "envelope".
- Bidders do not see each others' bids.
- Bids are collected by the auctioneer.
- The auctioneer determines the winner and the price to pay.
  - In the first-price sealed-bid auction, the highest bidder wins and pays his bid.

Vickrey (second-price sealed-bid) auction

- The highest bidder wins but only pays the *second highest bid* (or the *reserve price*, whichever is higher).
- Named after William Vickrey.
  - Awarded Nobel Prize in Economics in 1996.

Model

- Each bidder i has a **private valuation** (or, willingness to pay), θ_i, for the item.
- A bidder's utility is $u_i = \theta_i - p_i$ if he wins the item and has to pay the amount of p_i, and $u_i = 0$ otherwise.
  - Thus, θ_i is the price at which bidder i is indifferent between winning and losing.
  - The utility does not depend on the others' values.
- A bidder does not know the private valuations of other players, but he has beliefs about their distribution, described by the distribution function F or the corresponding probability density function f.
- Bidders maximise their expeced utility.
- We assume (for the moment only!) that they cannot collude.

Example: first-price auction

$$u_i(\theta_i) = \begin{cases} \theta_i - b_i & \text{if } \ b_i > \max_{j \neq i} b_j \\ 0 & \text{otherwise} \end{cases}$$

- Assume there are 2 players: $\theta_1 = 4$, $\theta_2 = 2$.
- Bids b_1, b_2 are integers from 0 to 4.
- Ties are broken randomly.
  - Write down the payoff matrix for this game.
  - Find the Nash equilibria.

Example: first-price auction

$\theta_2 = 2$

$\theta_1 = 4$

	0	1	2	3	4
0	2, 1	0, 1	0, 0	0, -1	0, -2
1	3, 0	1.5, 0.5	0, 0	0, -1	0, -2
2	2, 0	2, 0	1, 0	0, -1	0, -2
3	1, 0	1, 0	1, 0	0.5, -0.5	0, -2
4	0, 0	0, 0	0, 0	0, 0	0, -1

Example: first-price auction

$\theta_2 = 2$

$\theta_1 = 4$

	0	1	2	3	4
0	2, 1	0, 1	0, 0	0, -1	0, -2
1	3, 0	1.5, 0.5	0, 0	0, -1	0, -2
2	2, 0	2, 0	1, 0	0, -1	0, -2
3	1, 0	1, 0	1, 0	0.5, -0.5	0, -2
4	0, 0	0, 0	0, 0	0, 0	0, -1

Which aspect of the auction is missing?

Bayesian games

- Let $\theta_i \in \Theta_i$ denote the **type** of player i, and θ_{-i} denote the types of other players.
- In a Bayesian game,
  - a player's utility depends on its type θ_i and on the actions taken by all the players;
  - each player knows its own type θ_i, but does not know the type of other players θ_{-i};
  - players have a **prior distribution** over other players' types, which is *common knowledge*.

Bayesian games

A **Bayesian game** is a tuple $(N, (A_i, \Theta_i, u_i)_{i \in N}, Pr)$ where:

- N is a set of n agents;
- A_i is the set of actions available to player i;
- Θ_i is the type space of player i;
- $Pr : \Theta_1 \times \ldots \times \Theta_n \to [0, 1]$ is a common prior over types;
- $u_i : (\times_{i \in N} A_i) \times \Theta_i \to \mathbb{R}$ is the utility function for player i.

Bayesian games

- A strategy in a Bayesian game is a mapping $s_i : \Theta_i \to A_i$.
- We denote by $s_i(\theta_i) \in A_i$ the player i's action if his type is θ_i, and $s_{-i}(\theta_{-i})$ are the other players' actions given their types.
- The utility of player i depends on his type and the joint strategy of all agents: $u_i\left(\theta_i, s_i(\theta_i), s_{-i}(\theta_{-i})\right)$.
- The expected utility of player i is then:

$$\sum_{\theta_{-i}} Pr(\theta_{-i}) u_i\left(\theta_i, s_i(\theta_i), s_{-i}(\theta_{-i})\right)$$

Bayes-Nash equilibrium (BNE)

- A strategy profile is in **Bayes-Nash equilibrium** if for every $i \in N$, $\theta_i \in \Theta_i$ and $a_i \in A_i$:

$$\sum_{\theta_{-i}} Pr(\theta_{-i}) u_i\left(\theta_i, s_i(\theta_i), s_{-i}(\theta_{-i})\right) \geq \sum_{\theta_{-i}} Pr(\theta_{-i}) u_i\left(\theta_i, a_i, s_{-i}(\theta_{-i})\right)$$

Example: first-price sealed-bid

- It is easier to analyse the *continuous* form of auctions where the type space Θ is a continuum.
- This implies that the probability distribution is also continuous.
  - Let $\Theta = [0, 1]$ for each agent $i \in N$.
  - As before, $u_i(\theta_i) = \theta_i - b_i$ if $b_i > \max_{j \neq i} b_j$ and 0 otherwise.
  - The prior is i.i.d drawn from a uniform distribution $U(0, 1)$: let p.d.f $f(\theta) = 1$, $\theta \in [0, 1]$ and c.d.f $F(\theta) = \int_0^\theta f(\theta) \mathrm{d}\theta = \theta$.

Example: first-price sealed-bid

Theorem

The BNE strategy of player i is to bid $s_i(\theta_i) = \frac{n-1}{n}\theta_i$.

- Bidders shade their bid, while shading goes to 0 when $n \to \infty$.
- For $n = 2$ the BNE profile is $\left(\frac{1}{2}\theta_1, \frac{1}{2}\theta_2\right)$.
- The strategy is monotonically increasing:
  bidders with higher values bid higher.
- Thus, the bidder with the highest values always wins:
  the auction is said to be *efficient*.

First-price sealed-bid auction vs. Dutch auction

- In the first-price sealed-bid auction, a bidder wins the auction and pays her own bid if she submits the highest bid.
- In a Dutch auction, the first bidder who calls out that he will accept the current price wins the object at that price.
- Therefore, in both Dutch and first-price sealed-bid auctions, bidders are making tradeoffs between the probability of having the higest bid and the amount they pay if they win.

First-price sealed-bid auction vs. Dutch auction

- The set of strategies available to a bidder is the same in the Dutch auction (descending auction) as in the first-price sealed-bid auction, and therefore the Bayesian Nash Equilibrium bid functions are the same.
- Therefore, the Dutch auction is **strategically equivalent** to the first-price sealed-bid auction.
- This is why the descending bid auction is sometimes referred to as the *open first-price auction*.

Example: Vickrey auction

$$u_i(\theta_i) = \begin{cases} \theta_i - \max_{j \neq i} b_j & \text{if } \; b_i > \max_{j \neq i} b_j \\ 0 & \text{otherwise} \end{cases}$$

How should one bid here?

Revisit the earlier example:

- 2 players: $\theta_1 = 4$, $\theta_2 = 2$.
- Bids b_1, b_2 are integers from 0 to 4.
- Ties are broken randomly.

Example: Vickrey auction

$\theta_2 = 2$

$\theta_1 = 4$

	0	1	2	3	4
0	2, 1	0, 2	0, 2	0, 2	0, 2
1	4, 0	1.5, 0.5	0, 1	0, 1	0, 1
2	4, 0	3, 0	1, 0	0, 0	0, 0
3	4, 0	3, 0	2, 0	0.5, -0.5	0, -1
4	4, 0	3, 0	2, 0	1, 0	0, -1

Example: Vickrey auction

$\theta_2 = 2$

$\theta_1 = 4$

	0	1	2	3	4
0	2, 1	0, 2	0, 2	0, 2	0, 2
1	4, 0	1.5, 0.5	0, 1	0, 1	0, 1
2	4, 0	3, 0	1, 0	0, 0	0, 0
3	4, 0	3, 0	2, 0	0.5, -0.5	0, -1
4	4, 0	3, 0	2, 0	1, 0	0, -1

What's special about this payoff matrix?

Vickrey auction: strategy-proofness

- In the Vickrey auction, bidding $s_i(\theta) = \theta$ is a (weakly) dominant strategy.
- Such an auction is called
  - **truthful**,
  - or, equivalently, **strategy-proof**,
  - or **incentive-compatible in dominant strategies**.

Vickrey auction: strategy-proofness

Assume that a bidder bids its true valuation.

- If it finds that it has won the auction (i.e. he has submitted the highest bid), then the payment it makes is equal to the second highest bid.
- Hence, submitting a lower bid would have made no difference to this amount, but would simply have decreased the probability that it had won the auction in the first place.
- Likewise, submitting a higher bid would have made no difference to the outcome, nor the amount paid.

Vickrey auction: strategyproofness

Assume that a bidder bids its true valuation.

- If it is outbid by another bidder and so does not win the auction, then submitting a lower bid would have made no difference to the outcome.
- Submitting a higher bid, would have increased the probability that it won the auction.
- However, on winning it would discover that the second highest bid is greater than its true valuation (otherwise it would have won when it submitted its true valuation), and will thus make a loss.

Vickrey auction vs. English auction

- In a second-price sealed-bid (or, Vickrey) auction the person with the highest bid will win at a price equal to the second-highest bid.
- It is a dominant strategy for a bidder to bid its true value.
- Therefore, the bidder with the highest valuation wins the object and pays an amount equal to the value of the second highest bidder.

Vickrey auction vs. English auction

- In an English auction, an individual bidder will obtain a positive income from participating if and only if he can win the auction with a bid that is less than his valuation of the object.
- Therefore, a bidder drops out only when the bid "on the floor" exeeds his valuation.
- Hence, the bidder with the highest valuation wins the auction and pays an amount slightly higher than the value of the second highest bidder.
- This is why the English auction is sometimes referred to as the *open second-price auction*.

Revenue equivalence theorem

- Suppose valuations for the single item are drawn i.i.d from a continuous distribution over domain $[L, H]$ and agents are expected to be utility maximisers (risk neutral).
- Then any two auctions that:
  - In equilibrium always allocate the item to the bidder with the highest valuation, and
  - give an agent with valuation L an expected utility of 0,

  will give the same expected revenue to the auctioneer.
- This holds for all 4 auctions discussed, so they are **revenue equivalent**.

Collusion

- In an English auction, a group of bidders could form a bidding ring and agree not to bid against one another.
- By doing so, they can get the item at an artificially low price, and then decide how to allocate it amongst themselves.
- Individuals within the bidding ring have no incentive break the agreement, since such defection is visible to all bidders, and the bidder with the highest valuation can always respond by outbidding the defector and keeping the item for themselves.
- However, in a Dutch auction there is nothing to stop one member from breaking the agreement and winning the item (since the other bidders will have no opportunity to respond).

Collusion: example

- Consider collusion between two bidders, named 1 and 2, in a second price auction.
- Assume they tell each other their true values θ_1 and θ_2, and suppose $1 \geq \theta_1 > \theta_2 \geq 0$.
- Their goal is to set their bids in order to maximise the sum of their expected payoffs.
- If one of them wins, each player will get what he would have recieved had he bid honestly, and any surplus will be split evenly among them.
- What's the additional surplus the colluders can obtain and what are their optimal bids?

Collusion: example

Suppose these two players are the only two bidders in the game.

- In the second price auction, the winner w is the highest bidder who receives the payoff $b_w - p$, where $p \geq 0$ is the second highest bid. All other bidders receive a payoff of 0.
- For our problem, if one player bids nothing, then the other player, no matter what his bid is, will win the item for free.
- Since $\theta_1 > \theta_2$, it is obviously optimal for player 2 to bid nothing and for player 1 to bid any positive b_1. Then the sum of their payoffs is θ_1, the highest possible value.
- If they had bid honestly, player 1 would still have won the item, but paid a price of θ_2, for total payoff $\theta_1 - \theta_2$.
- So the extra surplus is θ_2 and each player gets $\frac{\theta_2}{2}$.

Collusion: example

Now suppose there is a third player who is not part of the collusion. Assume he draws his value independently from the uniform distribution on [0, 1].

- If we set $b_2 > 0$, there would be a positive chance that $\theta_3 < b_2$, and as a result player 1 will have utility $\theta_1 - b_2$.
- But if $\theta_3 > b_2$, then either 3 wins the item or player 1 wins the item at $\theta_1 - \theta_3$, which is independent of b_2.
- So setting $b_2 = 0$ is always the best or tied-for-best option.

Collusion: example

- Now, we are essentially playing a two-player second price auction where one player (3) is bidding honestly.
- So we know the best strategy for the other player (1) is to also bid honestly. Hence, $b_1 = \theta_1$.

Collusion: example

- Now, to get the surplus, we need to know the expected payoff for players 1 and 2 when they bid honestly.
- Since $\theta_1 > \theta_2$, player 2's payoff will always be zero when he is honest (he is always outbid by 1). Player 1's expected payoff is

$$\int_0^{\theta_2} (\theta_1 - \theta_2) d\theta_3 + \int_{\theta_2}^{\theta_1} (\theta_1 - \theta_3) d\theta_3 + \int_{\theta_1}^{1} 0 \; d\theta_3 = \frac{\theta_1^2 - \theta_2^2}{2}$$

Collusion: example

- Now, using the optimal collusion bids, player 1 will win whenever $\theta_3 < \theta_1$ and always pay price θ_3, so his expected payoff is
$$\int_0^{\theta_1} (\theta_1 - \theta_3) d\theta_3 = \frac{\theta_1^2}{2}$$
- The extra surplus is $\frac{\theta_2^2}{2}$, and each colluder gets $\frac{\theta_2^2}{4}$.

Cheating auctioneers

Some auction protocols are susceptible to manipulation on the part of the auctioneer:

- A dishonest auctioneer can exploit the Vickrey auction by lying about the 2nd highest bid.
- Shills can be introduced to inflate bidding prices in English auctions:
  - the auctioneer bids within his own auction.

Winner's curse

May occur in **common value auctions**
with incomplete information:

- There is a single valuation for the item.
- A bidder has an uncertain estimate of this valuation.

Since most auctions involve at least some amount of common value, and some degree of uncertainty about that common value, the winner's curse is an important phenomenon.

Winner's curse

The winner may overpay or be "cursed" in one of the two ways:

- The winning bid exceeds the value of the auctioned good such that the winner is worse off in absolute terms.
- The value of the good is less than the bidder expected, so he may still have a net gain but will be worse off than anticipated.

Example

Back in the 1950s, there was no accurate method to estimate the potential value of an offshore oil field.

- If, for example, an oil field had an actual intrinsic value of \$10 million, oil companies might guess its value to be anywhere from \$5 million to \$20 million.
- The company who wrongly estimated at \$20 million and placed a bid at that level would win the auction, and later find that it was not worth as much.

The term "winner's curse" was coined in the article *"Competitive bidding in high risk situations"* by Capen, Clapp and Campbell of the Atlantic Richfield Company (ARCO), published in the Journal of Petroleum Technology in 1971.

More examples

Other auctions where the winner's curse is significant:

- **Spectrum auctions**, in which companies bid on licenses to use portions of the electromagnetic spectrum.
  - The uncertainty would come from, for example, estimating the value of the cell phone market in a given area.
- **IPOs**, in which bidders need to estimate what the market value of a company's stock will be.
- **Pay per click advertising** online, where advertisers gain higher ranking if they bid higher amounts per click from a search engine user.

Uncertainty

- In a common value auction, the auctioned item is of roughly equal value to all bidders, but the bidders don't know the item's market value when they bid.
- Each player independently estimates the value of the item before bidding.

Overpay

- The winner of an auction is the bidder who submits the highest bid.
- Since the auctioned item is worth roughly the same to all bidders, they are distinguished only by their respective estimates of the market value.
- The winner, then, is the bidder making the highest estimate.
- If we assume that the average bid is accurate, then the highest bidder overestimates the item's value.
- Thus, the auction's winner is likely to overpay.

Overpay

Formally, this means that the **conditional expected value** of a bidder from the auction

- the expected value of the item, minus the expected price, conditioned on the assumption that the bidder wins the auction,

is negative.

Number of bidders

The severity of the curse grows with the number of bidders:

- This is because the more bidders, the more likely it is that some of them have overestimated the auctioned item's value.
- In technical terms, the winner's expected estimate is the value of the nth order statistic, which increases as the number of bidders increases.
- Hence, more bidders $\Rightarrow$ higher winner's curse.

Avoiding winner's curse

However, an actual overpayment will generally occur only if the winner fails to account for the winner's curse when bidding:

- an outcome that, according to the revenue equivalence theorem, need never occur.

Avoiding winner's curse

Savvy bidders will avoid the winner's curse by bid shading, or placing a bid that is below their *ex ante* estimation of the value of the item for sale—but equal to their *ex post* belief about the value of the item, given that they win the auction.

- The key point is that winning the auction is bad news about the value of the item for the winner.
- It means that he was the most optimistic and, if bidders are correct in their estimations on average, that he paid too much.
- Therefore savvy bidders revise their ex ante estimations downwards to take account of this effect.

(Single-good) auction variations

Reverse auctions, in which one buyer accepts bids from a set of sellers.

- This is what occurs when a buyer engages in a *request for quote*.
- Reverse auctions are simply auctions, in which we substitute the word "seller" for "buyer" and vice versa, and furthermore, negate all numbers indicating prices or bid amounts.

(Single-good) auction variations

Auctions with uncertain number of bidders, where the bidders are uncertain about the number of competitors they face.

- Often the case in sealed-bid auctions or in auctions held over the internet.
- Can also occur when there is an entry cost for participation in an auction, and some agents may just decide to stay away.

(Single-good) auction variations

Auctions with entry costs, in which the bidders have to pay a cost for participation in the auction.

- Interpretation of the entry-cost: the cost of researching an auction, driving (or navigating a Web browser) to it, and spending the time to bid.
- This can make equilibrium analysis much more complex.
  - Imagine that a first-price auction costs \$1 to attend.
  - How should bidders decide whether or not to attend, and then how should they decide to bid given that they're no longer sure how many other bidders will have chosen to attend?

Auctions with entry costs

Things are straightforward for second-price (or English) auctions:

- To decide whether to participate, bidders must evaluate their expected gain from participation.
- This means that the equilibrium strategy in these auctions does now depend on the distribution of other agents' valuations and on the number of these agents.
- The good news is that, once they have decided to bid, it remains an equilibrium for bidders to bid truthfully!

Auctions with entry costs

In first-price auctions (and, generally, other auctions that do not have a dominant- strategy equilibrium) auctions with entry costs are harder to analyze:

- The complication stems from the fact that a bidder is uncertain about other agents' valuations, and hence, is also uncertain about the number of agents who will decide to participate in the auctions.
- Since an agent's equilibrium strategy given that he has chosen to participate, depends on the number of other participants, the equilibrium strategy is more complicated to compute.
- That, in turn, makes it more difficult to determine the agent's expected gain from entering the auction in the first place.

Multi-unit auctions

- In practice there will often be more than one good to allocate, and different goods may end up going to different bidders.
- In **multiunit auctions**, there is only *one kind* of good available, but there are *multiple identical copies* of that good.
  - New cars, tickets to a concert, music downloads, shares of stock in the same company etc.

Sealed-bid multi-unit auctions

Overall, sealed-bid auctions in multiunit settings differ from their single-unit analogues in several ways.

- First, consider payment rules.
  - If, for example, there are three items for sale, and each of the top three bids requests a single unit, then each bid will win one good. How much each bidder should pay?
  - In the pay-your-bid scheme (the **discriminatory pricing rule**), each of the three top bidders pays a different amount, namely, his own bid.
  - Under the **uniform pricing rule** all winners pay the same amount; this is usually either the highest among the losing bids or the lowest among the winning bids.

Sealed-bid multi-unit auctions

Overall, sealed-bid auctions in multiunit settings differ from their single-unit analogues in several ways.

- Second, instead of placing a single bid, bidders generally have to provide a price offer for every number of units.
  - If a bidder simply names a number of units and is unwilling to accept any fewer, we say he has placed an **all-or-nothing** bid.
  - If he names one number of units but will accept any smaller number at the same price-per-unit we call the bid **divisible**.

English multi-unit auctions

- When moving to the multiunit case, designers of English auctions face all of the problems discussed above.
- However, since bidders can revise their offers from one round to the next, these auctions rarely ask bidders to specify more than one number of units along with their price offer.

Dutch multi-unit auctions

- In multiunit Dutch auctions, the seller calls out descending per-unit prices, and agents must augment their signals with the quantity they wish to buy.
- If that is not the entire available quantity, the auction continues.
  - The price can continue to descend from the current level,
  - can be reset to a set percentage above the current price,
  - or can be reset to the original high price.

Single-unit demand

- There are k identical goods for sale.
- The bidders want only 1 unit each, and have independent private values for these single units.

Single-unit demand

- The 2nd-price auction generalises to the $k+1^{\text{st}}$-price auction.
  - That is, the auction mechanism is to sell the units to the k highest bidders for the same price, and to set this price at the amount offered by the highest losing bid.
- The $k+1^{\text{st}}$-price auction is a special case of a mechanism called the **Vickrey-Clark-Groves** (VCG) mechanism, which applies to more general domains.
  - VCG is efficient and dominant-strategy truthful
    $\Rightarrow$ so is the $k+1^{\text{st}}$ auction.

Combinatorial auctions

- This is a broader auction setting, in which a whole variety of different goods are available in the same market.
- This differs from the multiunit setting because we no longer assume that goods are interchangeable.
- The key issue is that bidders' valuations depend strongly on which *set of goods* they receive.

Combinatorial auctions: examples

- Governmental auctions for the electromagnetic spectrum
- Energy auctions
- Corporate procurement auctions
- Auctions for paths (e.g., shipping rights; bandwidth) in a network, etc.

Combinatorial auctions: setting

- $N = \{1, \ldots, n\}$—set of bidders
- $G = \{1, \ldots, m\}$—set of bidders
- $(\theta_1, \ldots, \theta_n) \in \Theta^n$—agent types
- $v : 2^G \times \Theta \to \mathbb{R}$—valuation function

Combinatorial auctions: setting

A mechanism is a pair of functions (f, t), where:

- $f : \Theta^n \to X$—allocation function indicating what subsets of goods each agent receives, where $X \subseteq (2^G)^n$ is the set of feasible allocations;
- $t : \Theta^n \to \mathbb{R}^n$—transfer function indicating the payment receieved by each agent from the mechanism.
  - If negative, the agent pays to the mechanism.

The utility of agent i from allocation x depends linearly on the transfer (such a function is called *quasi-linear*):

$$u_i(x, t, \theta_i) = v(x, \theta_i) + t_i$$

VCG mechanism

Let $\hat{\theta}_i$ denote the type that agent i *reports* to the mechanism. In VCG,

- The allocation function f is *efficient*:

$$f(\hat{\theta}_1, \ldots, \hat{\theta}_n) \in \arg\max_{x \in X} \sum_{i \in N} v(x, \hat{\theta}_i)$$

- The transfer function pays each agent the amount by which his presence affects the other participants:

$$t_i(\hat{\theta}_1, \ldots, \hat{\theta}_n) = \sum_{j \neq i} v\left(f(\hat{\theta}_1, \ldots, \hat{\theta}_n), \hat{\theta}_j\right) - \max_{x \in X} \sum_{j \neq i} v(x, \hat{\theta}_j)$$

VCG mechanism

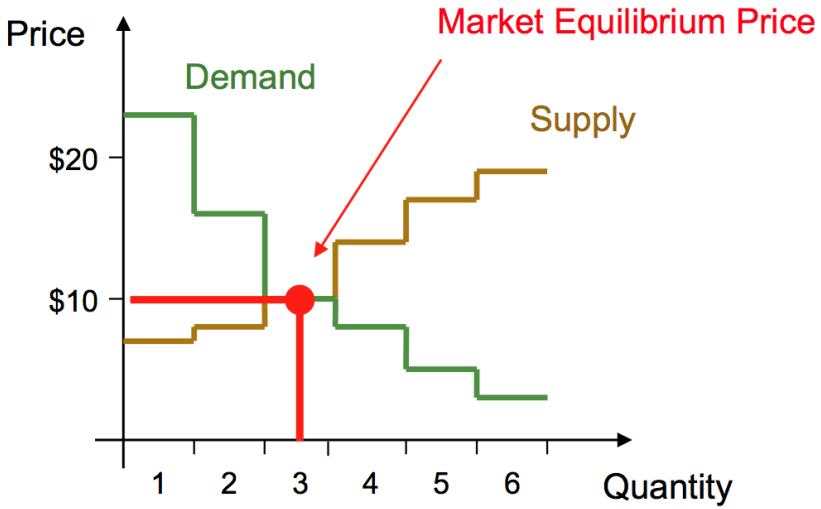
VCG mechanism is *efficient* and *truthful in dominant strategies*, and also:

- *individually rational*—no agent gets a negative utility;
- *weakly budget balanced*—the mechanism never pays the agents more than it collects from them.

Two-sided (double) auctions

- There are many buyers and sellers.
  - A typical example is the stock market.
- One of the primary models of such auctions are **continuous double auctions** (CDA), in which:
  - There are many potential buyers and sellers of many units of the same good.
  - Agents bid at their own pace and as many times as they want.

Continuous double auctions (CDA)



Trading at the market equilibrium is a Nash equilibrium.

THANK YOU!